What moves an open AI-edit detector on technical prose: controlled edits, one feature at a time

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Abstract

A correlation table of sentence scores suggests what to change in a text to lower its AI-edit score. We tested its five suggestions as controlled edits on 62 paragraphs of four of our robotics papers, scored with Pangram's open EditLens model, and three did not hold. Joining two sentences with a semicolon raised the score of 92 % of the paragraphs, by +0.047 on average, but an appended verdict sentence lowered it. Rewriting paragraphs against their sentence-level scores moved the four paper means from 0.211, 0.176, 0.202 and 0.233 to 0.153, 0.160, 0.189 and 0.207. In 39 human robot-learning papers from before November 2022 the paper means have a median of 0.071, and 4 of our four papers still score above all of them. Style features that go with low scores in the human papers did not lower the score of 5 of 7 paragraphs when we added them.

1 Introduction

Our four papers pass the slopscore and lmscan checks of the repository, and alphaXiv, which runs Pangram's detector, still flags them as AI-written. We wanted to know what in the text the detector reacts to, so that we could write the next paper without it.

We first read a table of 469 sentences from the four papers and compared the score contribution of each sentence with its features. Semicolons, contrast framing, advice, verdict words and colon set-ups went with higher scores, and "we", short sentences and citations went with lower ones. From this table we wrote rule 78 of the writing rules in road-to-robotics. A feature that goes with a high score can travel with something else, such as a long list of numbers. This paper changes one feature at a time and scores the result.

EditLens [5] takes a text and regresses how much an AI edited it, as a score between 0 and 1. Its authors trained it on pairs of a source text and an AI-edited version of it, from reviews, news, stories and web articles, edited with prompts such as "make it more descriptive" and "fix any mistakes". Our texts are experimental robotics papers, so we did not expect the same features to matter. The controlled edits below test that.

A score also needs a scale. We therefore scored 39 human robot-learning papers from before November 2022 in the same way, and we tried to write our paragraphs in the style that those papers suggest.

2 Setup

The scorer is `pangram/editlens_Llama-3.2-3B`, the open model, on a Jetson Orin NX at 25 W. The production detector that flagged the papers is a different and stronger model, and we did not run it. The model is licensed for non-commercial use and we used it on our own four papers only.

The texts are the rendered drafts of four papers: a register of re-tested robot-learning claims, a JEPA world model on a low-cost arm, reward models and a stopping rule, and chunking tricks for a VLA on an edge GPU. Citations are stripped and the model lower-cases its input. Paragraphs of at least 40 words are scored on their own, since shorter ones are noisy. There are 62 of them, in 469 sentences. As a human reference we took the first 40 results of an arXiv query for robot-learning papers submitted from 2018 to 2021 (the rule is in `repro/human-reference/PROTOCOL.md`) and read their PDFs with the same extractor we used on our own four built PDFs. 1 paper had too few scoreable paragraphs and was dropped. Reference lists and acknowledgements that the extractor had kept as prose (116 paragraphs, found after the first scoring because they were among the highest scores) were excluded, which leaves 39 papers and 1,365 paragraphs.

An edit changes one thing in every paragraph that contains it, and we report the mean change in the paragraph score with a 95 % bootstrap interval over paragraphs. An edit that changes only case and spacing gives exactly 0.000, because the model normalizes both. We removed semicolons and colons from the texts before this test, so for these two we could only test adding them.

*Code, result files and the checks that regenerate every number: `github.com/ravediamond/paper-editlens-sensitivity`. The texts under test belong to four other papers of the author; their rewrite history is in `repro/rewrite-loop/`.

edit	paragraphs changed	change in score (95 % CI)
join sentence pairs with a semicolon	62	+0.047 (+0.039 to +0.056)
append a generic closer, "The result holds on this rig."	62	+0.017 (+0.012 to +0.022)
append a contrast closer, "a limit of the setting, not of the method"	62	+0.011 (+0.006 to +0.017)
append advice, "Check the simple baseline first."	62	+0.001 (−0.004 to +0.005)
append a verdict, "The row fails."	62	−0.015 (−0.020 to −0.011)
append a sentence about the rig, "We ran this on the Jetson Orin NX."	62	−0.001 (−0.007 to +0.004)
append a real sentence from another paragraph of the same paper	62	+0.003 (−0.005 to +0.011)
change "we" and "our" to "I" and "my"	52	+0.001 (−0.004 to +0.006)
shuffle the sentences of a paragraph	62	+0.000 (−0.012 to +0.012)
drop the parentheticals	27	−0.018 (−0.033 to −0.005)
split at ", and" and ", but"	28	−0.014 (−0.020 to −0.007)
replace every number with "many"	55	−0.021 (−0.030 to −0.012)
keep only the first half of the sentences	62	−0.029 (−0.044 to −0.015)

Table 1: Controlled edits on the original paragraphs. A positive change means the edited paragraph scores higher, which is worse for us.

feature	correlation table	controlled edit
semicolons	+0.028	+0.047, holds
contrast framing	+0.029	+0.011, holds and is smaller
advice or imperatives	+0.021	+0.001, does not hold
verdict words	+0.020	−0.015, reversed
"we" or "our" doing something	−0.020	+0.001, does not hold

Table 2: The sentence-level correlations we first used as rules, against the controlled edit of the same feature.

3 Results

3.1 Controlled edits

The semicolon and the two closers raise the score, and the verdict sentence lowers it. Advice, the sentence about the rig, a real sentence of the same paper and the pronoun change do nothing, so adding length or changing the pronoun does not move the score. Shuffling does nothing either. Dropping the parentheticals, splitting at "and" and "but", and masking numbers lower the score. Keeping half of a paragraph lowers it most.

3.2 What the earlier table got right

Two of five features held. The verdict result is the one we did not expect. In the papers a verdict word sits in sentences that also carry a number, a bar and a citation, which may be what raised the score in the table. We did not test that. A verdict sentence appended on its own does not raise it.

paper	paragraphs	mean before	mean after
register of re-tested claims	23	0.211	0.153
JEPA world model	22	0.176	0.160
reward models and a stopping rule	21	0.202	0.189
chunking tricks on an edge GPU	18	0.233	0.207

Table 3: Word-weighted mean paragraph score before and after the rewrite.

change	paragraph	before	after
name the works in place of "papers propose several uses"	JEPA introduction	0.315	0.164
split one list sentence into plain statements	register, cold reruns	0.253	0.139
open with the AUROC example and give the class letter after it	register, classes B and C	0.253	0.145

Table 4: The three largest drops in one paragraph, by the change that caused them.

3.3 The rewrite loop

We rewrote paragraphs from the highest-scoring sentences down. Each round scored a few candidate rewrites with the placeholders of the paper filled in, and kept the lowest one that preserved every number and claim. In the register paper one sentence moved to another paragraph and one that repeated another was dropped. There were 90 candidates in 50 paragraph-rounds. 50 of the candidates scored lower than the text they replaced, and the best candidate of a round improved the paragraph in 64 % of the rounds, by 0.032 on average.

The largest single drops came from three changes, listed in Table 4. Merging sentences into longer ones with "so" or "which means" raised a limitations paragraph from 0.275 to 0.365 in the first variant, and the other two variants did the same. Paragraphs made of numbers, bars and verdicts stayed above the median of human results paragraphs (0.045) whatever we did.

3.4 A lexicon of words

We also tried to turn the sentence scores into a list of words. A lexicon built on three papers predicts the sentence effects of the fourth with a correlation of 0.116, and the 95th percentile of shuffled labels is 0.096. Deleting single words in the 20 highest paragraphs (2,042 deletions) agrees in sign with the sentence source for 48 of 80 words. Only 19 words pass our strict test. Words that name a method for talking about itself (protocol, rule, committed, reports, CI) go with higher scores, and plain task words and units go with lower ones. The effect of one word is small, since the mean deletion changes a paragraph by +0.013 and the largest by 0.157. The list

section type	paragraphs	median	90th percentile
introduction	167	0.053	0.177
related work	114	0.063	0.153
method	133	0.073	0.174
results	140	0.045	0.169
conclusion	82	0.043	0.116
other or not recognised	689	0.069	0.187

Table 5: Paragraph scores of the human reference papers by section type, read from the PDFs. Section types come from the headings and are rough.

feature	rank correlation with score	low	high	ours
numeric tokens	−0.12	4.29	3.66	8.56
passive constructions	−0.12	1.12	0.89	0.40
contrast words	−0.10	0.36	0.35	0.08
connective words	−0.13	0.29	0.12	0.21
commas	+0.09	5.16	5.99	4.35
words per paragraph	−0.28	199	125	137

Table 6: Style features of the human paragraphs by score, and of ours, read from the PDFs. Counts are per 100 words except the last row. Low is the 542 human paragraphs below 0.05 and high the 211 above 0.15.

is a tie-breaker between synonyms.

3.5 A human reference

Table 5 gives the paragraph scores of the 39 human papers by the type of section. The paper means run from 0.033 to 0.123 with a median of 0.071. The median paragraph scores 0.063, 15 % of the paragraphs score above 0.15, and 5 % score above 0.25. Results paragraphs have a median of 0.045, so a results paragraph at 0.15, the level where ours stopped, is high for a human one.

Read the same way, our four built PDFs have paper means of 0.182, 0.130, 0.145 and 0.138, and 56 % of their 50 scoreable paragraphs score above 0.15. All four lie above the human median, and 4 of the 4 lie above the highest human paper.

3.6 Writing in the style of the human papers

Our papers score above the human ones, so we measured style features of the human paragraphs and of ours (Table 6) and looked for the ones that go with a low score. The rank correlations are weak, at most 0.30 in absolute value. The largest gaps between our paragraphs and the human ones are the numeric tokens, the passive constructions, and the contrast and connective words. Our paragraphs have 8.56 numeric tokens per 100 words where the human paragraphs below 0.05 have 4.29.

We rewrote 7 paragraphs of two of our papers in that direction, in our own words and with every number kept, joining clauses with contrast and connective words and using the passive for procedures. Of the 8 candidates, 5 scored higher than the paragraph they replaced and 2 lower, and the mean change was +0.020. No candidate lowered 5 of the 7 paragraphs. The related-work paragraph of the register paper rose from 0.194 to 0.280. The features that go with low scores across human paragraphs did not lower ours when we added them, which is the same gap between a correlation and an edit as in Section 3.2.

4 Limitations

The model is the open 3B one, and the detector that flagged the papers is another. We did not run the production model, so none of these effects is shown to carry over to it. Running it on the four papers costs about 3.2 dollars for each full pass at the listed API rate, and we did not spend it.

We drafted the four papers with Claude Code, and Claude Code wrote the rewrites in this paper too. The author has not yet read the rewrites paragraph by paragraph. The open EditLens model scores a paragraph low when it sees little AI editing in it. We do not read the score as who wrote the paragraph.

Table 1 changes one feature per edit. A real rewrite changes several at once. We have not run the test that applies the rewrite moves of Section 3.3 one at a time to the same paragraphs. That test would show whether their effects add up. All 62 paragraphs come from the four papers of one author, on one SO-101 arm and one Jetson Orin NX. The five appended sentences are our own examples, and other wording could behave differently. The human reference is the first 40 results of one query, not a sample of the field, and its section types are rough because many headings were not recognised. Our own four PDFs give only 50 scoreable paragraphs, so their section types are not compared. The style trial is 7 paragraphs from two papers, one round, with features we chose, so it shows that these edits did not help in that sample and does not rule them out.

5 Related work

EditLens [5] introduces the continuous score and reports binary and ternary detection results, and we use its open 3B model. Its authors state that the model has a non-zero error rate and can cause harm when used to accuse someone. The AI Polish study [4], which EditLens evaluates on, shows that binary detectors flag lightly polished text as AI. Work on paraphrasers and humanizers [2, 1, 3] studies how they degrade the detection of AI text. We read only the descriptions of these in the EditLens paper and not the works themselves. We start from human-directed drafts of technical prose and ask which edits move an open model's score, and we report the edits that did nothing.

6 Conclusion

Of the five features we had read off a correlation table, a semicolon and a contrast closer raised the score and the other three did not. A suspected feature should therefore be tested as an edit before it becomes a writing rule. Our four papers score above the human median, and our result paragraphs stayed above the median of human results paragraphs. Adding the style features of the human papers did not lower our scores. In our rewrites, paragraphs with named things and plain "we ran X on Y" sentences scored lower than paragraphs of procedure words. We have not tested whether the rest of the gap is content that only the author can add, such as the reasons for a decision.

References

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